

MCEN90053

Industrial Systems and Simulation

Assignment 2

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Beverage Manufacturing Process for Profit Maximisation

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Contents

1	Introduction	1
2	Model Development Process	1
2.1	Strategy Outline	1
2.2	Demand Estimation Procedure	1
2.2.1	The Uses and Focuses of Three Forecasting Methods	1
2.2.2	Justification for choice of Weighted Moving Average Method	3
3	Anylogic Model	4
3.1	Using the AnyLogic interface to model the factory procedure	5
3.2	Verification and Validation of the AnyLogic Model	5
3.3	Model Insights and Limitations	5
3.3.1	Insights of the factory system from the model	5
4	Production Game	6
4.1	Using the model to inform investments	6
4.2	How to choose the manufacturing configuration	6
4.2.1	Breakdown of weekly strategy	7
4.3	Evaluating positive and negative strategy outcomes	8
4.4	Points of improvement of the model	8
5	Production Game Review	8
5.1	Model benefits (and working performance)	9
5.2	Possible improvements to the model for better performance	9
5.2.1	Parameter setting adjustments in prediction	9
5.2.2	Dynamic production scheduling adjustments	9
5.2.3	Improving supply chain responsiveness and inventory management	10
5.2.4	Automation streamlining of procedure	10
5.3	Uncertainty of Demand Forecasting	10
5.3.1	Complexity of market demand fluctuations	10
5.3.2	Limitations of predictive modelling	11
5.3.3	Delayed effects in production and supply chains	11
5.3.4	Uncontrollable factors in the external environment	11
6	Conclusion	11
A	Appendix	12
A.1	30-week period simulation results	12
A.2	Demand forecasting plotted data	18
A.2.1	Moving average method	18
A.3	Agent-based simulation: Weekly Analysis Template	18
A.4	Determination of bottleneck	18

TEAM MEMBER CONTRIBUTIONS

Name and ID	Contribution
Karan Gaur (1587591)	Worked through the bottleneck procedure. Produced and finalised the Excel spreadsheets for calculations relating to profit and inventory, for integration with the AnyLogic code. Determined the strengths and weaknesses of the model, both in terms of operation and concept.
Yung Jian Hu (1084099)	Wrote the introduction, described the AnyLogic model to be used on the game day. Wrote about the validation and verification sections. Insights of the model were also considered, such as the important information which can be uncovered due to the model. Limitations of the model were also discussed. A rough strategy analysis was also considered, including an explanation on the way the model was used to inform the whole production process. Analysed and confirmed the bottleneck determination. For the production game review, talked about the working performance of the model, and reflected on the performance on the simulation day.
Qiurui Qi (1582658)	Provided detailed explanations of the three forecasting methods, including the presentation of calculation results. Discussed the suitability of the chosen method in relation to the game strategy, and its alignment with the AnyLogic model. Analysed its overall impact during the pre-game phase, and explained why other methods were not suitable.
Zuifeng Song (1238344)	Uncertainties in demand forecasting are analysed, pointing out the impacts of market demand fluctuations, forecasting model limitations, production and supply chain response delays and uncontrollable factors in the external environment on the accuracy of demand forecasting. Suggestions for model performance improvement are also presented, including dynamic parameter adjustment, introduction of intelligent scheduling algorithms, and enhancement of supply chain responsiveness.

1 Introduction

Factory process management is a highly important factor in the context of systems engineering. The effective identification of bottlenecks, as well as clarity in anticipating the fluctuations in demand, are crucial for maximising profit. This report highlights a discrete event simulation (DES)-based approach to determine the optimum strategy for profit maximisation of a beverage manufacturing facility over a 30-week period.

The facility produces three types of beverages: A, B and C, which go through a set of manufacturing procedures. These include, but are not limited to, the mixing of the drink, the bottling of the mixed drink, and finally packaging into designated boxes. Therefore, the objective of the DES procedure to be implemented is to ensure that the production timeline strategy delivers products in a timely manner, so that costs associated with warehouse storage is minimised.

By analysing the performance of the model in response to different fluctuating demand patterns, the bottlenecks in the entire manufacturing procedure are identified to achieve the aforementioned profit goals. Furthermore, the demand forecasting procedure conducted as part of the analysis is detailed here.

2 Model Development Process

2.1 Strategy Outline

Using the AnyLogic model we designed our aim was to maximise the profitability by strategically managing the production flows across the various product types and the different bottle sizes. By utilising the AnyLogic software, bottlenecks can be detected, and production can be dynamically adjusted, based on the real-time monitoring of the warehouse stocks and profitability, based on weekly demand.

2.2 Demand Estimation Procedure

Demand forecasting analysis aims to offer production planning effectively to maximise profits over 30 weeks. Given the demand fluctuations and limited project resources, the method chosen for demand estimation was the “weighted moving average” method to support our game strategy. This method is suitable for our own bottleneck that we have discussed earlier in the report, and it also adapts to the dynamic demand changes on the AnyLogic model. In that case, it will reduce warehouse costs. This section describes three different forecasting methods in detail, as well as provide a justification for selecting the “weighted moving average” method, by illustrating how it supports subsequent operations.

2.2.1 The Uses and Focuses of Three Forecasting Methods

(1) Moving Average Method: The moving average method works best in stable demand scenarios because it has to average data over multiple periods to smooth out short term fluctuations. However, as shown in Figure 1 and Table 1 (in Appendix A.2.1), the forecast remains constant at 46666.7, failing to capture demand changes. As the prediction range (28187.9 to 65145.4) is excessively wide, the method does not prove its ability to reflect accurate shifts. That may lead to inventory surpluses or shortages due to delayed adjustments. Consequently, the moving average method falls short in providing the support for flexible production planning and dynamic demand management needed in this game.

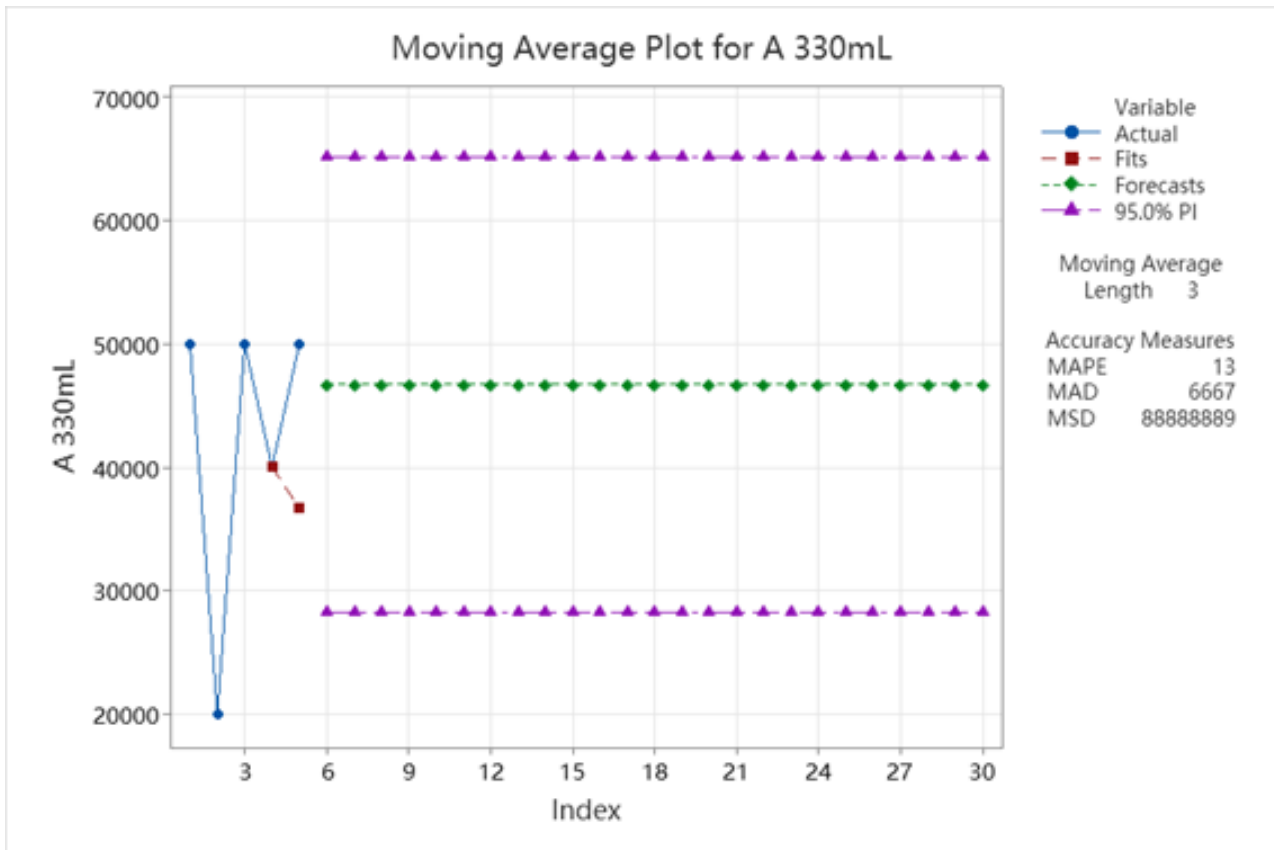


Figure 1: Moving average plot for Beverage A 330 mL

(2) Weighted Moving Average: The weighted moving average method pays more attention to the recent data, making it more responsive to shifts in demand. As this project entails frequent demand changes, especially within a five-week period, this method is suitable for the quick adjustments to the production schedule. With this method, we can focus on producing products that exhibit the properties of high-demand and high-profit. This can be integrated with the AnyLogic model seamlessly (through on-the-spot data input into AnyLogic). This adaptability aligns with the objective of improving production efficiency by reducing inventory costs and increasing the turnover rate.

(3) Exponential Smoothing: The exponential smoothing method is suitable for short-term demand fluctuations without significant trends or seasonal changes because it quickly responds by assigning higher weights to recent data. However, in our project, we face frequent demand fluctuations and short production cycles with alternating high and low demand. In both the 3-week and 5-week forecast windows, the limitation of a fixed α value prevents this method from adjusting flexibly to these fluctuations, leading to overly stable results instead of capturing the variability. Adjusting the α value alone does not resolve this issue, as changes in the forecast window are not fully captured, resulting in poor handling of complex demand variations. Additionally, this method has a delayed response in recognising long-term trends. This delay limits the ability to adjust production line priorities dynamically. Additionally, the method restricts flexible resource allocation. These limitations make it difficult to manage bottlenecks and quickly adapt resource schedules.

As a result, inventory surpluses or shortages may occur to a greater extent. This increase warehousing costs and reduces system efficiency. Therefore, the exponential smoothing method is not able to provide the flexible production and dynamic adjustment standards required. The forecast results in the figure show insufficient variability and responsiveness to changes.

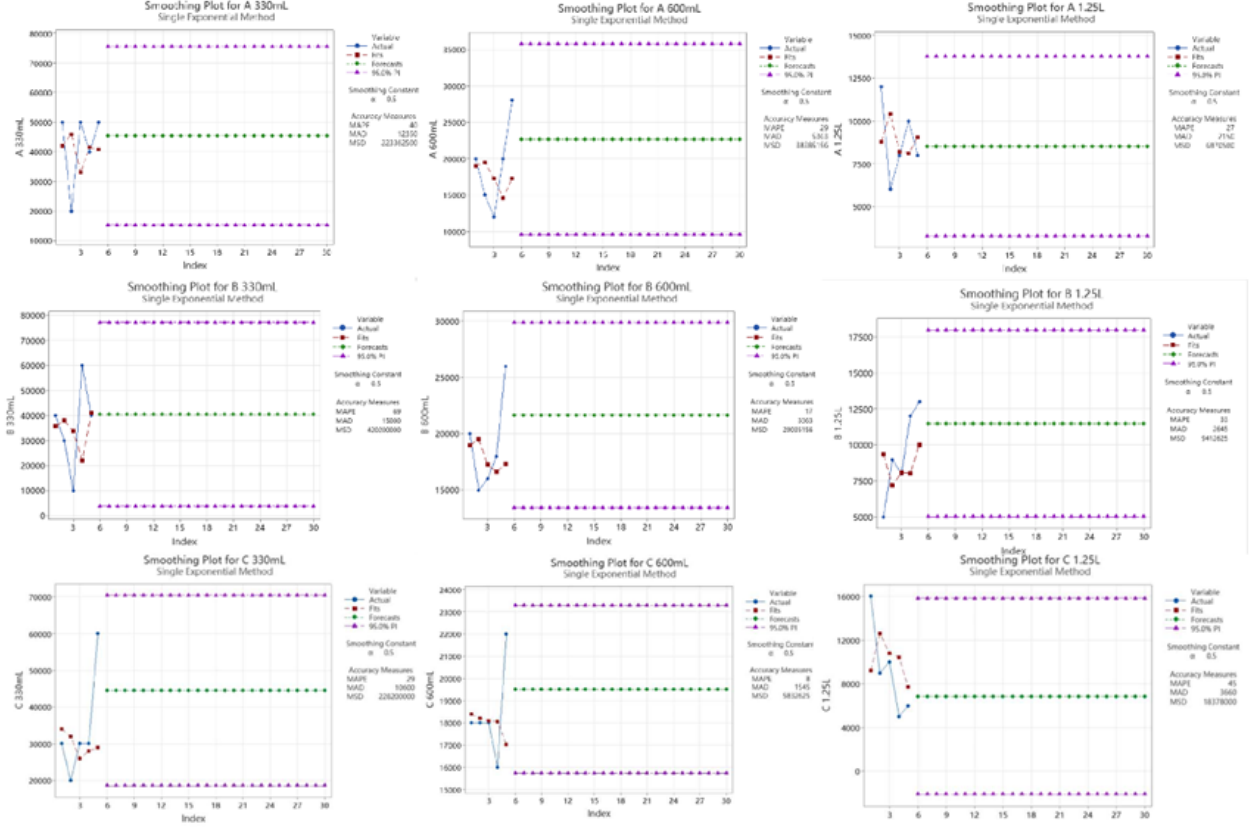


Figure 2: Exponential smoothing forecast plot for various bottle sizes.

2.2.2 Justification for choice of Weighted Moving Average Method

To thoroughly analyse the necessity of choosing the weighted average moving method, we can illustrate this based on the chosen strategy and the characteristics of the AnyLogic model from two main aspects.

Accuracy and Timeliness of the strategy: The project strategy focuses on adjusting production priorities across different bottle sizes to improve efficiency and boost profits. Early in the project, demand was especially unpredictable. The trend of Product Group A shows the fluctuations within the first three weeks. To make sure that high-profit items are produced and the inventory is effectively managed, the method required should be capable of responding quickly to recent demand changes. Unlike the moving average method and exponential smoothing method, the weighted average method puts more emphasis on the latest data, making it more adaptable and precise in capturing short-term changes. The result informs the forecasting procedure, which helps align production decisions and resource allocation with market trends. Additionally, the risk of excess inventory from delayed forecasting is reduced, which supports the goal of profit maximisation.

Compatibility with the AnyLogic Model: Our AnyLogic model is designed to replicate complex production stages like mixing, filling, and packaging, while bottlenecks can impact capacity and flexibility. To support dynamic scheduling and optimise resources, the weighted moving average method offers quick, updated demand forecasts. This helps make real-time changes to production plans and prioritise tasks. The method's timely forecasts feed directly into the model, aiding in simulating production loads and tracking warehouse costs. This

ensures the model responds effectively to bottlenecks and production shifts. By using this approach, resource allocation can be improved, delays can be minimised, and inventory pressures can be relieved – overall boosting the model’s performance.

Ultimately, the choice of the weighted moving average method aligns perfectly with our strategy’s requirements for precise, flexible responses to demand fluctuations and the dynamic adjustment capabilities of the AnyLogic model. This method not only improves the accuracy and timeliness of forecasts but also supports efficient production decision-making and resource optimisation in a complex production environment, allowing the project to respond to market fluctuations more effectively.

3 Anylogic Model

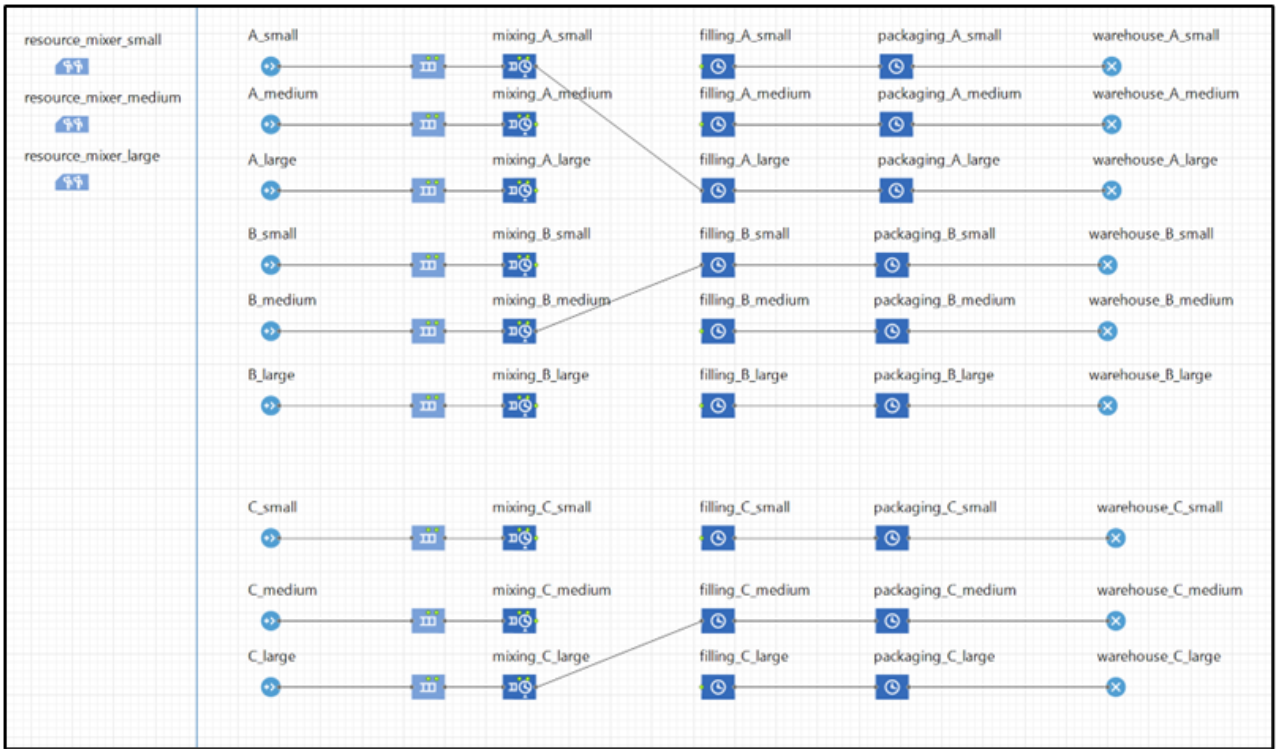


Figure 3: The AnyLogic model used for the simulation.

For simplicity, the model was designed with nine parallel processes, operating in the same way, for each of the drinks produced. When running the model, it is made clear which processes are active, to ensure the simulation is accurate (in other words, there must not be any duplicate usage of the mixer or the filler for a particular configuration of drink). The model includes three production lines for each of the three beverages, corresponding to the size of the packaging as well as the mixing tank that is being used.

Depending on the choice of configuration for a certain production week, a manual change is made to the connection between the mixer and the filler. This signifies which mixing line to use for the particular beverage in question and sets the finished drinks to be filled in a particular size. As the agents proceed through the simulation, they will pass through their designated sink. The number of units that pass through the sink are taken to be the final quantity of the factory’s production that week.

The inputs to the model are the estimated demand values from the demand forecasting methodology, discussed in Section 2.2. Note that a conversion must be made from the number

of demanded units to the number of agents in the AnyLogic model, prior to the simulation.

3.1 Using the AnyLogic interface to model the factory procedure

1. Assume that a demand forecast has been generated. The output of the demand forecast is a number of bottles of a particular drink and a particular size.
2. Convert the number of bottles to the number of litres, by multiplying by the number of bottles with the number of litres per bottle.
3. Convert the number of litres per bottle to the number of agents, for simulation in the AnyLogic model *One agent is 10 litres*.
4. Input the number of agents into the corresponding source. For example, if it is required to use the small mixer to make Beverage A, and to bottle Beverage A into large bottles of 1.25 L, connect a line between the service for the small mixer and the delay for the large filling tank. This means that all the agents will proceed through the small mixer, and get led to the large filling tank, where all mathematical calculations are pre-set in the AnyLogic blocks regarding the filling time per agent.
5. Determine the number of agents that proceed to the sink of that production line. Due to the three mixing tanks and three packaging options, there should only be three sinks that have a nonzero number of agents arriving.
6. The output from the previous step can be converted again from the number of agents to the number of bottles. This depends on the size of the bottle and can be calculated by dividing the number of agents by the size of the bottle. The overall number of bottles can be compared with the required minimum lot size to determine whether or not a product can be delivered. This calculation is performed externally through an Excel sheet.

3.2 Verification and Validation of the AnyLogic Model

To verify the completeness of the model, it was tested with a sample source arrival rate. Note that the source arrival rates correspond to the desired production output, which was estimated from the external demand forecasting step. As the model runs, it does not incur any errors in the system. In other words, AnyLogic was able to fully process the model and provide a result based off a set of inputs. Because of this, the model can be considered to be verified.

However, validation is a separate procedure which must be undertaken. This procedure ensure that the output of the functional (verified) model is acceptable and provides sensible results for use in future analysis.

There are multiple ways to perform the validation. The validation of the model may be done through benchmarking the different demand forecasting methods to determine which one best captures the varying nature of the demand. One method performed consistently with the demand estimation goals (Weighted Moving Average Method), which adds to the model's credibility. Additionally, a sensitivity analysis may be performed on some of the demand estimation parameters to assess the stability of the model's demand estimation, where reasonable stability may suggest more reliable results.

3.3 Model Insights and Limitations

3.3.1 Insights of the factory system from the model

One of these assumptions relate to the fact that the agents follow a linear process through the AnyLogic processes (sources, delays, services, and sinks). Therefore, their behaviour is limited to the workings of these processes. This leads to certain peculiarities within the model – one of which is visible through the mixing stage (figure: mixing peculiarity). It can be seen that the

top two ports of the service block in AnyLogic that corresponds to the mixing process show two numbers. The number on the top left signifies the number of agents in the queueing part of the service, and the number on the top right shows the number of agents currently being mixed.

The above peculiarity is noteworthy since the entire AnyLogic simulation runs on a timescale of weeks. In other words, each run of the simulation corresponds to one week of production in the factory. At the end of the week, the simulation stops running. This means that any agents currently still moving through either queues or services in the main line do not proceed any further, and their current position is shown in the output model.

The AnyLogic simulation model which we designed as shown in Figure 3 models the production process of the Beverage company from mixing, to filling and finally to the packing. This allows us to do the testing for various scenarios each week and make informed decisions. One of the insights that we infer from the simulation is that filling process acts as a bottleneck, constraining the overall production capacity. This is also shown from calculations in Appendix A.4 that demonstrate the limiting nature of the filling stage. Despite making the investments to streamline other stages, the speed of the filling process continues to act as a hindrance and impedes the flow of production. This bottleneck necessitates a strategy where we prioritise smaller, faster-to-fill bottles, especially during the initial stages.

4 Production Game

4.1 Using the model to inform investments

There is a distinct weekly profit margin for each of the 9 products, which impacts production priorities. This occurs due to the real-time monitoring method where an analysis of the profit and warehouse stock are performed each week according to varying demand. For example, if 600 mL of Beverage B yielded the highest profit margin for a given week, then production on that product is prioritised to maximise the profitability. By following this approach, the production process is varied on a weekly basis (according to game rules), in an effort to reduce priority on less profitable products.

Warehouse stock management is another highly important aspect of the strategy. Excessive warehouse stocks increase holding costs, which damage profits. Overproduction is avoided, by adjusting production according to current stock levels while analysing profitability. By analysing if the stock levels have reached critical thresholds, production can be reduced for that particular product. (*Note:* Mathematical analysis can be performed here according to statistical standard deviations of inventory levels to determine the critical thresholds.)

Regarding investments in exchange for technical improvements, there is the availability to: (1) reduce the changeover time of a mixing barrel when transitioning between the production of one drink and another, or; (2) reduce the fluctuation in production rates at both the mixing and filling processes. As the analysis showed that a bottleneck existed in the filling step, a decision was reached to direct investments into the filling process, further outlined in Section 4.2.1, when discussing the strategy over the entire 30 week period.

4.2 How to choose the manufacturing configuration

The manufacturing configuration was calculated externally through the use of an Excel spreadsheet. The method adopted emphasised the importance of the changing demand, and its effect on overall profit. Answering this question requires substituting another question in place of it: based off a certain demand for the week: (1) Which mixers should be used for which drinks, and (2) what sizes should those drinks be packaged in?

Profit Analysis Week 4											
Bottle type Type	Bottle Type	Demand	Product Cost	Production Cost	Warehouse Cost/product	Warehouse Cost	Bottle type Type	Total Cost	Product Price	Selling Price	Profit
A	330 ml	45000	0.9	40500	1	45000	A	40500	1.5	67500	27000
A	600 mL	28000	1.6	44800	2	56000	A	44800	3	84000	39200
A	1.25 L	13000	2	26000	3	39000	A	26000	4	52000	26000
B	330 ml	20000	0.7	14000	1	20000	B	14000	1.2	24000	10000
B	600 mL	24000	1.1	26400	2	48000	B	26400	2.3	55200	28800
B	1.25 L	8000	1.5	12000	3	24000	B	12000	3.5	28000	16000
C	330 ml	65000	1.1	71500	1	65000	C	71500	1.8	117000	45500
C	600 mL	34000	2	68000	2	68000	C	68000	3.5	119000	51000
C	1.25 L	13000	3	39000	3	39000	C	39000	4.5	58500	19500

Figure 4: Profit analysis of Week 4 of a particular week, from Excel.

The most efficient way to obtain an answer to these questions is to simulate the production of all 9 possible configurations of drink mixing and packaging. Then, to determine the most profitable allocation of resources, the configuration which yields the maximum overall profit is chosen.

A screenshot of the excel table is provided in Figure 4, and shows the all the possibilities for profits, depending on the mixing and packaging pattern. Choosing the configuration which gives the maximum profit, that configuration is run on the AnyLogic model, according to the following steps.

Figure 4 shows high-priority products (Focus on Higher Margins) based on the demand for a particular week is given as:

- Product C, 600 mL: Priority due to its high profit margin and moderate demand level. This bottle size provides a good balance of profitability and relatively lower filling time compared to larger bottles.
- Product C, 330 mL: Priority as it offers a high profit margin and reasonable demand. While the larger size may require more resources, the profit potential justifies the allocation.
- Product A, 1.25 L: Prioritize production of this bottle size due to its high profit margin, despite its lower demand. Focus on producing enough to meet demand, as its high profit margin will significantly boost profitability.

4.2.1 Breakdown of weekly strategy

A phased approach that takes into account stock levels, demand, and profitability. Below is a breakdown of our week-by-week strategy:

- Weeks 1-10: Initial focus is on high-demand products to build up inventory, particularly for sizes with higher profitability. Investments are directed toward optimizing the filling process for this product to address the bottleneck.
- Weeks 11-20: Shift the focus to Products with high profitability along with the warehouse stocks that the particular stocks have and make additional investments aimed at reducing filling times (e.g., from 10 hours to 5 hours) to increase capacity for this product. This helps alleviate the impact of the filling bottleneck on the rest of the system.
- Weeks 21-30: Implement a balanced strategy that priorities fulfilling all product orders, with an emphasis on minimising warehouse costs by keeping stock levels manageable. Production is balanced to prevent overstocking and ensure optimal resource allocation.

4.3 Evaluating positive and negative strategy outcomes

There are two main contributing components to the overall strategy. One component (Strategy Component 1) is the method of deciding which configuration of mixing and filling are appropriate for the current week. Another (Strategy Component 2) is the method of forecasting the demand. These two methods go hand-in-hand in influencing the overall factory management procedure. Therefore, they are equally influential to the overall profit generated.

Deciding the mixing and filling configuration: A positive outcome of Strategy Component 1 refers to its sensitivity to changing demand. Other strategies in this component may include committing to the most profitable combination of drink and size, and fully focusing production on that drink and size. However, this method does not consider the fact that there may be some weeks which see no demand for that particular product. In effect, strategists who adopt this methodology may be stuck in periods where their factory earns zero profit whatsoever. The strategy adopted by this team in considering the demand allows the production to be flexible according to the demand forecasting strategy, which is a significant contributor to profit earned.

Forecasting the demand: A positive outcome regarding demand forecasting is that the demand forecasting method has a flexibility to choose between a three or five-week moving average. A three-week average is more beneficial in that it has a shorter reaction time to demand. These faster adjustments mean that there is less of a risk of over or underproduction, leading to warehouse costs or insufficient stock. Ultimately, for this scheme, only the most recent three weeks out of the data from 5 weeks are fed into the estimation scheme to determine the appropriate actions for future production scheduling.

Furthermore, this may lead to more frequent changeovers in production, leading to less overall time spent on production. On the other hand, a five-week moving average scheme does not encounter the same detriments. As the data input time window is longer for a five-week scheme compared to a three-week scheme, it does not suffer from the same operational disruptions for shorter schemes. This leads to a more stable planning procedure overall. The longer time window reduces the number of changeovers and therefore improves the overall number of units produced. However, the opposite downside can be seen – this production scheme increases the risk of incurring more warehouse costs due to overproduction.

4.4 Points of improvement of the model

A glaring potential source of difficulty comes from the large interconnectedness of the simulation procedure. Due to the high reliance on individual components of the model (demand estimation to production scheduling to AnyLogic model output to profit calculation), the model is highly reliant on smooth operation over all of these components. As mentioned above, the model stresses simplicity in individual components (the Excel spreadsheets and the AnyLogic model used are straightforward without complicated programming). However, this introduces new challenges for the modelling team, since the interconnectedness of the team must sufficiently compensate the lack of automation in the overall procedure.

5 Production Game Review

After the running the model through the 30-week period, the performance of the model is evaluated. In particular, both the successful and problematic aspects of the model are considered, as well as the overall profit generated through the demand prediction procedure. These con-

siderations can be classified into either of three categories: (1) Speed rating (turnaround time from data acquisition to model output and weekly results); (2) Coherence of model operation while obtaining weekly results; and (3) Total profit obtained (effectiveness of the production strategy).

5.1 Model benefits (and working performance)

The model was initially designed to prioritise simplicity. In other words, there were not many complicated features with the coding or the model development. This is a design consideration that did not initially consider the high speed required for the production game. Therefore, when trialling the model in these more realistic conditions, the whole period of the 30 weeks was not able to be simulated in the available working time. Note that this data was obtained after the game, and a completed set of results for the entire 30-week period was calculated and attached in Appendix ??

The main advantage of this Real-Time Monitoring strategy is its adaptability. By continuously tracking profits and warehouse stock levels, we can quickly adjust production priorities to meet weekly demands and maximise profits. Combined with the AnyLogic simulation model, this approach provides valuable insights into the the production status, allowing us to test different configurations and make informed decisions. This is made possible by incorporating an agent-based modelling into our strategy to align with our simulation parameters. A screenshot of the weekly analysis template is shown in Appendix A.3.

Additionally, there not many errors from AnyLogic. The few minor malfunctions which occurred purely did so because of accidental input errors (which were easily fixed). However, the processing of data due to Excel encountered more difficulties. The simulation requires changing units multiple times, as the AnyLogic model works in units of 1 agent to 10 litres of beverage.

5.2 Possible improvements to the model for better performance

In the whole process of demand forecasting, production scheduling and supply chain management, although the current model has a certain demand response and production optimisation capability, there is still much room for improvement in order to cope with the complex and changing market demand and the influence of the external environment. In order to further improve the model performance and response capability, improvements can be made in the following aspects.

5.2.1 Parameter setting adjustments in prediction

Firstly, To improve the accuracy of demand forecasting, the model should optimise its parameter settings. Currently, the weighted moving average and exponential smoothing methods rely on fixed parameters, such as the value of the smoothing coefficient α , which limits the model's flexibility when demand fluctuates frequently. Dynamic parameter adjustment methods, such as genetic algorithms or Bayesian optimisation, could enable the automatic adjustment of parameter values in response to real-time demand changes. This would enhance the model's forecast accuracy and adaptability, making it more responsive to complex demand patterns.

5.2.2 Dynamic production scheduling adjustments

Within production scheduling, the scheduling mechanism of the current model is a static decision derived from the forecast results, which does not account for the cost and time delay problems that may arise from the frequent switching of production lines. The production scheduling

strategy can be improved by adding a real-time production adjustment mechanism that dynamically adjusts the production priorities whenever demand is expected to change, hence minimising the efficiency loss and cost increase that arise from frequent switching. Furthermore, the AnyLogic Model can introduce rule-based intelligent scheduling algorithms or even potentially machine learning-driven scheduling optimisation that enables the system to flexibly allocate resources and change production sequences according to demand fluctuations and the current production status, thus providing a better response to uncertain demand changes. Note that these changes are highly complicated, but can be tackled with adequate time and knowledge.

5.2.3 Improving supply chain responsiveness and inventory management

Another key determinant of the performance of a model is associated with slower supply-chain response times, which affects, above all, the time-lag between production output modification and inventory adjustment undertaken. To respond in advance to demand fluctuations and thus reduce the risks associated with mismatched supply and demand, forecasting supply lead time and setting safety stocks can be absorbed by supply-chain management. Moreover, we are entitled to deploy the model with these further means-utilising Economic Order Quantity (EOQ) and safety stock-to enable the manufacturing plants to become more flexible in a rapid response to demand shift to avoid added costs and management pressure associated with surplus or shortage of inventories.

5.2.4 Automation streamlining of procedure

The current model in respect of data processing and integration depends on Excel sheets, which are handling manual data conversion and processing that can be blown by manual errors and inefficiency. Automating this entire piece of interface systems with a centralised point of resource management shall ensure direct conversion of demand forecast data to production plans, minimising human interventions thereby enhancing efficiency.

5.3 Uncertainty of Demand Forecasting

Uncertainty in demand forecasting is a key challenge in production and supply chain management, especially in this beverage production plant project, where the team needed to make accurate production decisions under conditions of frequent demand fluctuations. This uncertainty comes from: (1) Complexity of market demand fluctuations; (2) Limitations of predictive modelling; (3) Delayed effects in production and supply chains; and (4) Uncontrollable factors in the external environment.

5.3.1 Complexity of market demand fluctuations

Fluctuations in demand for beverage products do not always exhibit predictable cyclical or seasonal patterns. Consumer demand may be affected by the external environment, seasonal changes, market trends, and unexpected events. Therefore, it is difficult to capture the sudden changes in demand, even when demand forecasts are made from historical data. In other words, variability can never be eradicated. The difficulty of forecasting is further exacerbated by the fact that demand changes are not entirely consistent across products and packages over a 30-week simulation period.

5.3.2 Limitations of predictive modelling

Although we chose the weighted moving average method as the main forecasting method, which is able to give higher weights to recent data to enhance responsiveness to changes in demand, this method still has limitations in the case of highly volatile demand. This method can better reflect short-term demand changes, but it requires appropriate weight settings. In reality, it is difficult to set the weights to perfectly fit the demand changes at each stage, which leads to the possibility that the forecast results may not accurately reflect the demand fluctuations.

In addition, the weighted moving average method is deficient in capturing long-term trends and seasonal fluctuations in demand. Short-term forecast windows allow for a quick response to changes in demand but lead to frequent production switching and higher adjustment costs, while longer forecast windows, while more stable, are less sensitive to recent changes and are prone to overstocking or shortages, which in turn increase storage and dispatch costs.

5.3.3 Delayed effects in production and supply chains

Relying on warehouse stock levels as a control mechanism may lead to overproduction if the stocks are not well managed. This will increase holding costs which damage profits. In a real-world scenario, excess stocks may need to be discounted and cleared to regain warehouse space and avoid increased holding costs.

5.3.4 Uncontrollable factors in the external environment

In addition to demand fluctuations within the market, uncontrollable factors in the external environment are an important source of influence on demand forecasts. For example, policy changes, raw material price fluctuations, supply chain disruptions or unforeseen social events can have an impact on demand that is difficult to incorporate into traditional forecasting models.

Agent-based simulation mentions that the model can consider certain external environmental factors, but in real demand forecasting, these uncontrollable factors are often sudden and can have an unexpected impact on market demand patterns. As a result of these uncontrollable factors, even forecasts based on the best historical data and model parameters may deviate significantly from actual demand.

6 Conclusion

The project and preceding report highlight the use of discrete event simulation (DES) and the demand forecasting methodology to optimise the production of a beverage manufacturing plant over a 30-week time period. With a rapid response to changes in demand using the AnyLogic platform and weighted moving average methodology, the model reduces costs related to stock-holding and increases production efficiency. The synergy between demand forecasting and DES offered solid support for balancing production volumes against delays, enabling the plant to become more flexible and to cope with dynamic market demand. Though results are encouraged, the model was constrained in its sensitivity to parameters, the flexibility of scheduling-a compromise with responsiveness to the supply chain. Future improvement of the model with enhanced responsiveness and accuracy may be achieved through dynamic parameter tuning, real-time scheduling optimisation and full automation of data processing.

Hence in conclusion this project exhibits that the combination of DES and Demand Forecasting in Manufacturing optimisation is effective and provides solid grounds for insiders of production decisions in complex market environments. This approach, despite inherent drawbacks of simulation, yields higher operational efficiency but provides a stronger support for

manufacturing firms to maximise profits.

A Appendix

A.1 30-week period simulation results

(shown on the following five pages for all five periods)

Period 1

Week 1

Product	Size	Order	Produced
A	330mL	75000	0
A	600mL	30000	38334
A	1.25L	18000	23200
B	330mL	20000	0
B	600mL	24000	0
B	1.25L	14000	0
C	330mL	70000	90910
C	600mL	24000	0
C	1.25L	20000	0
Weekly Profit			43822

Week 2

Product	Size	Order	Produced
A	330mL	40000	0
A	600mL	22000	0
A	1.25L	14000	17600
B	330mL	40000	0
B	600mL	14000	0
B	1.25L	6000	0
C	330mL	45000	57576
C	600mL	24000	31250
C	1.25L	5000	0
Weekly Profit			37624

Week 3

Product	Size	Order	Produced
A	330mL	45000	0
A	600mL	12000	0
A	1.25L	17000	21600
B	330mL	80000	0
B	600mL	16000	0
B	1.25L	13000	0
C	330mL	75000	96970
C	600mL	26000	33334
C	1.25L	10000	0
Weekly Profit			35062

Week 4

Product	Size	Order	Produced
A	330mL	45000	0
A	600mL	28000	0
A	1.25L	13000	16800
B	330mL	20000	0
B	600mL	24000	0
B	1.25L	8000	0
C	330mL	65000	78788
C	600mL	34000	42500
C	1.25L	13000	0
Weekly Profit			50312

Week 5

Product	Size	Order	Produced
A	330mL	50000	0
A	600mL	36000	45000
A	1.25L	14000	0
B	330mL	50000	0
B	600mL	38000	0
B	1.25L	4000	0
C	330mL	60000	75758
C	600mL	14000	0
C	1.25L	19000	23920
Weekly Profit			62382

Overall Profit 229202

Period 2

Week 6

Product	Size	Order	Produced
A	330mL	50000	60607
A	600mL	16000	16667
A	1.25L	7000	0
B	330mL	65000	0
B	600mL	14000	0
B	1.25L	5000	0
C	330mL	15000	0
C	600mL	10000	0
C	1.25L	19000	24000
Weekly Profit			33959

Week 7

Product	Size	Order	Produced
A	330mL	55000	60607
A	600mL	32000	33334
A	1.25L	14000	0
B	330mL	25000	0
B	600mL	12000	0
B	1.25L	15000	16000
C	330mL	45000	0
C	600mL	14000	0
C	1.25L	4000	0
Weekly Profit			44525

Week 8

Product	Size	Order	Produced
A	330mL	60000	0
A	600mL	20000	0
A	1.25L	7000	0
B	330mL	45000	0
B	600mL	26000	0
B	1.25L	16000	16800
C	330mL	70000	78788
C	600mL	30000	28334
C	1.25L	8000	0
Weekly Profit			13144

Week 9

Product	Size	Order	Produced
A	330mL	45000	0
A	600mL	10000	0
A	1.25L	5000	0
B	330mL	65000	60607
B	600mL	16000	0
B	1.25L	16000	17600
C	330mL	25000	0
C	600mL	28000	33334
C	1.25L	11000	0
Weekly Profit			-2075

Week 10

Product	Size	Order	Produced
A	330mL	80000	96970
A	600mL	32000	40000
A	1.25L	15000	19200
B	330mL	20000	0
B	600mL	26000	0
B	1.25L	16000	0
C	330mL	55000	0
C	600mL	24000	0
C	1.25L	9000	0
Weekly Profit			77230

Overall Profit 166783

Period 3

Week 11

Product	Size	Order	Produced
A	330mL	55000	63637
A	600mL	28000	0
A	1.25L	8000	0
B	330mL	80000	0
B	600mL	22000	0
B	1.25L	8000	0
C	330mL	25000	0
C	600mL	32000	26667
C	1.25L	9000	6400
Weekly Profit			-48177

Week 12

Product	Size	Order	Produced
A	330mL	50000	0
A	600mL	12000	0
A	1.25L	16000	19200
B	330mL	50000	0
B	600mL	18000	0
B	1.25L	7000	0
C	330mL	45000	54546
C	600mL	34000	41667
C	1.25L	5000	0
Weekly Profit			80322

Week 13

Product	Size	Order	Produced
A	330mL	55000	0
A	600mL	32000	35000
A	1.25L	6000	0
B	330mL	45000	0
B	600mL	12000	0
B	1.25L	18000	20800
C	330mL	70000	84849
C	600mL	22000	0
C	1.25L	6000	0
Weekly Profit			100551

Week 14

Product	Size	Order	Produced
A	330mL	25000	0
A	600mL	30000	0
A	1.25L	6000	0
B	330mL	65000	0
B	600mL	18000	0
B	1.25L	8000	0
C	330mL	45000	51516
C	600mL	28000	30000
C	1.25L	11000	9600
Weekly Profit			-34184

Week 15

Product	Size	Order	Produced
A	330mL	75000	0
A	600mL	26000	0
A	1.25L	12000	14400
B	330mL	70000	0
B	600mL	36000	0
B	1.25L	4000	0
C	330mL	80000	100000
C	600mL	36000	43334
C	1.25L	4000	0
Weekly Profit			92312

Overall Profit 190867

Period 4

Week 16

Product	Size	Order	Produced
A	330mL	35000	0
A	600mL	14000	0
A	1.25L	13000	0
B	330mL	50000	0
B	600mL	36000	41667
B	1.25L	16000	16000
C	330mL	60000	51516
C	600mL	16000	0
C	1.25L	14000	0
Weekly Profit			12350

Week 17

Product	Size	Order	Produced
A	330mL	35000	0
A	600mL	20000	0
A	1.25L	5000	0
B	330mL	50000	0
B	600mL	36000	45000
B	1.25L	4000	0
C	330mL	75000	87879
C	600mL	24000	0
C	1.25L	9000	8000
Weekly Profit			40821

Week 18

Product	Size	Order	Produced
A	330mL	55000	0
A	600mL	38000	43334
A	1.25L	17000	0
B	330mL	60000	0
B	600mL	30000	0
B	1.25L	17000	20000
C	330mL	55000	54546
C	600mL	16000	0
C	1.25L	4000	0
Weekly Profit			12986

Week 19

Product	Size	Order	Produced
A	330mL	55000	0
A	600mL	10000	0
A	1.25L	14000	0
B	330mL	80000	90910
B	600mL	38000	38334
B	1.25L	19000	21600
C	330mL	55000	0
C	600mL	26000	0
C	1.25L	14000	0
Weekly Profit			54222

Week 20

Product	Size	Order	Produced
A	330mL	25000	0
A	600mL	34000	40000
A	1.25L	5000	0
B	330mL	75000	0
B	600mL	34000	38334
B	1.25L	10000	0
C	330mL	35000	0
C	600mL	22000	0
C	1.25L	16000	15200
Weekly Profit			22132

Overall Profit 142511

Period 5

Week 21

Product	Size	Order	Produced
A	330mL	65000	72728
A	600mL	32000	0
A	1.25L	9000	0
B	330mL	40000	0
B	600mL	18000	0
B	1.25L	19000	20800
C	330mL	35000	0
C	600mL	34000	33334
C	1.25L	16000	0
Weekly Profit			-2796

Week 22

Product	Size	Order	Produced
A	330mL	60000	0
A	600mL	32000	0
A	1.25L	13000	0
B	330mL	60000	0
B	600mL	40000	50000
B	1.25L	19000	22400
C	330mL	80000	87879
C	600mL	26000	0
C	1.25L	15000	0
Weekly Profit			103921

Week 23

Product	Size	Order	Produced
A	330mL	30000	0
A	600mL	16000	0
A	1.25L	17000	16000
B	330mL	40000	30304
B	600mL	22000	0
B	1.25L	12000	0
C	330mL	15000	0
C	600mL	40000	45000
C	1.25L	12000	0
Weekly Profit			-28304

Week 24

Product	Size	Order	Produced
A	330mL	20000	0
A	600mL	38000	0
A	1.25L	18000	20000
B	330mL	55000	0
B	600mL	38000	0
B	1.25L	13000	0
C	330mL	65000	69697
C	600mL	38000	40000
C	1.25L	19000	0
Weekly Profit			123803

Week 25

Product	Size	Order	Produced
A	330mL	80000	84849
A	600mL	18000	0
A	1.25L	13000	16000
B	330mL	40000	0
B	600mL	28000	0
B	1.25L	5000	0
C	330mL	40000	0
C	600mL	32000	31667
C	1.25L	5000	0
Weekly Profit			-3183

Overall Profit 193441

A.2 Demand forecasting plotted data

A.2.1 Moving average method

Table 1: Forecasts for the moving average method.

Period	Forecast	Lower	Upper	Period	Forecast	Lower	Upper
6	46666.7	28187.9	65145.4	19	46666.7	28187.9	65145.4
7	46666.7	28187.9	65145.4	20	46666.7	28187.9	65145.4
8	46666.7	28187.9	65145.4	21	46666.7	28187.9	65145.4
9	46666.7	28187.9	65145.4	22	46666.7	28187.9	65145.4
10	46666.7	28187.9	65145.4	23	46666.7	28187.9	65145.4
11	46666.7	28187.9	65145.4	24	46666.7	28187.9	65145.4
12	46666.7	28187.9	65145.4	25	46666.7	28187.9	65145.4
13	46666.7	28187.9	65145.4	26	46666.7	28187.9	65145.4
14	46666.7	28187.9	65145.4	27	46666.7	28187.9	65145.4
15	46666.7	28187.9	65145.4	28	46666.7	28187.9	65145.4
16	46666.7	28187.9	65145.4	29	46666.7	28187.9	65145.4
17	46666.7	28187.9	65145.4	30	46666.7	28187.9	65145.4
18	46666.7	28187.9	65145.4				

A.3 Agent-based simulation: Weekly Analysis Template

Analysis Week																		
Bottle	Capacity	Demand(Bottles)	Updated Demand	Lot size	No. of Lots	Rounding up	Production cost	Total Cost	Product Price	Total sales	Profit	Demand(Litres)	Agents(Input)	Agent(Output)	reduced Units(Litres)	reduced Units(Bottle)	reduced Units(Bottles)	Rounding up
A	0.33		0	6000	0	0	0.9	0	15	0	0	0	0	0	0	0	0	0
A	0.6		0	3240	0	0	16	0	3	0	0	0	0	0	0	0	0	0
A	1.25		0	1560	0	0	2	0	4	0	0	0	0	0	0	0	0	0
B	0.33		0	6000	0	0	0.7	0	12	0	0	0	0	0	0	0	0	0
B	0.6		0	3240	0	0	11	0	2.3	0	0	0	0	0	0	0	0	0
B	1.25		0	1560	0	0	15	0	3.5	0	0	0	0	0	0	0	0	0
C	0.33		0	6000	0	0	11	0	18	0	0	0	0	0	0	0	0	0
C	0.6		0	3240	0	0	2	0	3.5	0	0	0	0	0	0	0	0	0
C	1.25		0	1560	0	0	3	0	4.5	0	0	0	0	0	0	0	0	0

Figure 5: The weekly analysis template for calculations on Excel is shown here.

A.4 Determination of bottleneck

Tables 2 and 3 show the filling process as being the bottleneck, and hence the limiting factor, in the factory's operation. It can be seen that, by calculating the maximum capacity of each stage and normalising the result to bottles per week, the filling stage consistently shows the lowest result for all products.

Table 2: **Process Times and Rates**

Category	Details	Time/Rate
Mixing Times	2000L tank	2-2.5 hours
	4000L tank	3-3.5 hours
	6000L tank	4-4.5 hours
Filling Rates	330 mL bottles	25-35 bottles/min
	600 mL bottles	12-18 bottles/min
	1.25 L bottles	6-9 bottles/min
Change-over Times	Filling change-over	10 hours per product change
	Mixing change-over	Unspecified
Packaging Rates	330 mL bottles	6000 lots in 30 minutes
	600 mL bottles	3240 lots in 15 minutes
	1.25 L bottles	1560 lots in 7 minutes

Table 3: **Production Capacity Calculations.**

Stage	Details	Calculation	Max Capacity
Mixing	2000L tank (330 mL)	$\frac{50 \text{ hrs/week}}{2.25 \text{ hrs/batch}} \times 6060 \frac{\text{bottles}}{\text{batch}}$	$\approx 133,320 \frac{\text{bottles}}{\text{week}}$
	4000L tank (600 mL)	$\frac{50 \text{ hrs/week}}{3.25 \text{ hrs/batch}} \times 6666 \frac{\text{bottles}}{\text{batch}}$	$\approx 99,990 \frac{\text{bottles}}{\text{week}}$
	6000L tank (1.25 L)	$\frac{50 \text{ hrs/week}}{4.25 \text{ hrs/batch}} \times 4800 \frac{\text{bottles}}{\text{batch}}$	$\approx 57,600 \frac{\text{bottles}}{\text{week}}$
Filling	330 mL line	$30 \frac{\text{bottles}}{\text{min}} \times 3000 \frac{\text{min}}{\text{week}}$	$= 90,000 \frac{\text{bottles}}{\text{week}}$
	600 mL line	$15 \frac{\text{bottles}}{\text{min}} \times 3000 \frac{\text{min}}{\text{week}}$	$= 45,000 \frac{\text{bottles}}{\text{week}}$
	1.25 L line	$7.5 \frac{\text{bottles}}{\text{min}} \times 3000 \frac{\text{min}}{\text{week}}$	$= 22,500 \frac{\text{bottles}}{\text{week}}$
Packaging	330 mL	$\frac{50 \text{ hrs/week}}{0.5 \text{ hrs/lot}} \times 6000$	$= 600,000 \frac{\text{bottles}}{\text{week}}$
	600 mL	$\frac{50 \text{ hrs/week}}{0.25 \text{ hrs/lot}} \times 3240$	$= 648,000 \frac{\text{bottles}}{\text{week}}$
	1.25 L	$\frac{50 \text{ hrs/week}}{0.117 \text{ hrs/lot}} \times 1560$	$= 663,000 \frac{\text{bottles}}{\text{week}}$